Theorem 1. *If x and n are positive reals,*

$$0 < \frac{\log(x)^2}{n} < \epsilon, \quad \textit{and}$$

$$y = (1 + \log(x)/n)^n,$$

then

$$(1 - \epsilon)\, x < y < x < (1 + \epsilon)\, y.$$

(i.e. $x \approx y$)

Corollary 2. *If x and n are positive reals,*

$$0 < \frac{\log(x)^2}{n} < 0.0001, \quad \textit{and}$$

$$y = (1 + \log(x)/n)^n,$$

then

$$0.9999\, x < y < x < 1.0001\, y.$$

(i.e. $x \approx y$)

Proof. Using for all $u > 0$ and all real a,

$$u - u^2/2 < \log(1 + u) < u, \quad \text{and} \tag{1}$$

$$\exp(-a) > 1 - a, \tag{2}$$

we have

$$\begin{aligned}
y &= (1 + \log(x)/n)^n \\
\log(y) &= n \log(1 + \log(x)/n) \\
n\left(\log(x)/n - \frac{\log(x)^2}{2n^2}\right) &< \log(y) < \log(x) \\
\log(x) - \frac{\log(x)^2}{2n} &< \log(y) < \log(x) \\
x \exp\left(-\frac{\log(x)^2}{2n}\right) &< y < x \\
x\left(1 - \frac{\log(x)^2}{2n}\right) &< y < x \\
x(1 - 0.0001) &< y < x \\
0.9999\, x &< y < x \\
y < x &< y/0.9999 \\
y < x &< 1.0001y
\end{aligned}$$

So,

$$0.9999\, x < y < x < 1.0001y.$$

□